1. Find eigenvalues and corresponding eigenvectors of the following matrices:

(a)

$$\begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}, \quad \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}.$$

(i) $Au = \lambda u = \lambda I u \quad \therefore (A - \lambda I) u = 0$

$\therefore \det(A - \lambda I) = 0$

$\because A - \lambda I = \begin{bmatrix} 3-\lambda & 4 \\ -1 & 7-\lambda \end{bmatrix}$

$\therefore (3-\lambda)(7-\lambda) + 4 = \lambda^2 - 10\lambda + 25 = (\lambda - 5)^2 = 0$

$\therefore \lambda_1 = \lambda_2 = 5$

$\therefore \begin{bmatrix} -2 & 4 \\ -1 & 2 \end{bmatrix} u = 0 \quad \therefore u = \begin{bmatrix} 2 \\ 1 \end{bmatrix} t \ (t \neq 0)$

(ii) $A - \lambda I = \begin{bmatrix} 1-\lambda & 3 \\ 4 & 2-\lambda \end{bmatrix}$

$\therefore (1-\lambda)(2-\lambda) - 12 = \lambda^2 - 3\lambda - 10 = (\lambda - 5)(\lambda + 2)$

1. Find eigenvalues and corresponding eigenvectors of the following matrices:

(a)

$$\begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 3 \\ 4 & 2 \end{pmatrix}.$$

$\therefore \lambda_1 = 5,\ \lambda_2 = -2$

For $\lambda_1 = 5$, $\begin{bmatrix} -4 & 3 \\ 4 & -3 \end{bmatrix} u_1 = 0$ $\qquad \therefore u_1 = \begin{bmatrix} 3 \\ 4 \end{bmatrix} p$

For $\lambda_2 = 5$, $\begin{bmatrix} 3 & 3 \\ 4 & 4 \end{bmatrix} u_2 = 0$ $\qquad \therefore u_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} q$

$(p, q \neq 0)$

1. Find eigenvalues and corresponding eigenvectors of the following matrices:

(b)

$$\begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix}, \qquad \begin{pmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{pmatrix}.$$

(i) $A - \lambda I = \begin{bmatrix} 1-\lambda & 2 & 1 \\ 6 & -1-\lambda & 0 \\ -1 & -2 & -1-\lambda \end{bmatrix}$

$$\therefore \det(A-\lambda I) = -6\begin{bmatrix} 2 & 1 \\ -2 & -1-\lambda \end{bmatrix} + (-1-\lambda)\begin{bmatrix} 1-\lambda & 1 \\ -1 & -1-\lambda \end{bmatrix} - 0$$

$$= -6(-2\lambda) + (-1-\lambda)(\lambda^2)$$

$$= -\lambda^3 - \lambda^2 + 12\lambda = -\lambda(\lambda-3)(\lambda+4)$$

$\therefore \lambda_1 = 0,\ \lambda_2 = 3,\ \lambda_3 = -4$

For $\lambda_1 = 0$, $\begin{bmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{bmatrix} u_1 = 0$

$\therefore \begin{bmatrix} 1 & 2 & 1 \\ 0 & -13 & -6 \\ 0 & 0 & 0 \end{bmatrix} u_1 = 0 \qquad \therefore u_1 = \begin{bmatrix} -\frac{1}{13} \\ -\frac{6}{13} \\ 1 \end{bmatrix} t = \begin{bmatrix} -1 \\ -6 \\ 13 \end{bmatrix} t$

$(t \neq 0)$

1. Find eigenvalues and corresponding eigenvectors of the following matrices:

(b)

$$\begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix}, \qquad \begin{pmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{pmatrix}.$$

For $\lambda_2 = 3$, $\begin{bmatrix} -2 & 2 & 1 \\ 6 & -4 & 0 \\ -1 & -2 & -4 \end{bmatrix} u_2 = 0$

$\therefore \begin{bmatrix} -2 & 2 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} u_2 = 0 \qquad \therefore u_2 = \begin{bmatrix} -1 \\ -\frac{3}{2} \\ 1 \end{bmatrix} q = \begin{bmatrix} 2 \\ 3 \\ -2 \end{bmatrix} q$

$(q \neq 0)$

For $\lambda_3 = -4$, $\begin{bmatrix} 5 & 2 & 1 \\ 6 & 3 & 0 \\ -1 & -2 & 3 \end{bmatrix} u_3 = 0$

$\therefore \begin{bmatrix} 5 & 2 & 1 \\ 0 & \frac{3}{5} & -\frac{6}{5} \\ 0 & 0 & 0 \end{bmatrix} u_3 = 0 \qquad \therefore u_3 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} q \quad (q \neq 0)$

1. Find eigenvalues and corresponding eigenvectors of the following matrices:

(b)

$$\begin{pmatrix} 1 & 2 & 1 \\ 6 & -1 & 0 \\ -1 & -2 & -1 \end{pmatrix}, \qquad \begin{pmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{pmatrix}.$$

(ii) $\det \begin{bmatrix} 4-\lambda & 0 & 1 \\ 2 & 3-\lambda & 2 \\ 1 & 0 & 4-\lambda \end{bmatrix} = 0$

$\therefore\ 0 + (3-\lambda)\begin{vmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{vmatrix} + 0 = (3-\lambda)(\lambda^2 - 8\lambda + 15)$

$= (3-\lambda)(\lambda-3)(\lambda-5) = 0$

$\therefore\ \lambda_1 = \lambda_2 = 3,\ \lambda_3 = 5$

For $\lambda_1 = \lambda_2 = 3$,

$\begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 2 \\ 1 & 0 & 1 \end{bmatrix} u_{1,2} = 0 \qquad \therefore \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} u_{1,2} = 0$

$\therefore\ u_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} p \ (p \neq 0),\ u_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} q \ (q \neq 0)$

For $\lambda_3 = 5$,

$\begin{bmatrix} -1 & 0 & 1 \\ 2 & -2 & 2 \\ 1 & 0 & -1 \end{bmatrix} u_3 = 0 \qquad \therefore \begin{bmatrix} -1 & 0 & 1 \\ 0 & -2 & 4 \\ 0 & 0 & 0 \end{bmatrix} u_3 = 0 \qquad \therefore\ u_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} t \ (t \neq 0)$

2. Check if the following matrices are diagonalisable:

$$\begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad \begin{pmatrix} 0 & -2 & -3 \\ 1 & 3 & 3 \\ 0 & 0 & 1 \end{pmatrix}.$$

① $\det \begin{bmatrix} 3-\lambda & 1 & 0 \\ 0 & 3-\lambda & 0 \\ 0 & 0 & 4-\lambda \end{bmatrix} = 0$

$\therefore (3-\lambda)^2(4-\lambda) = 0$ $\qquad \therefore \lambda_1 = \lambda_2 = 3,\ \lambda_3 = 4$

For $\lambda_1 = \lambda_2 = 3$,

$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} u_{1,2} = 0$ $\qquad \therefore u_{1,2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} t \ (t \neq 0)$

$\therefore$ nullity $= 1 <$ number of repeated $\lambda = 2$

$\therefore$ Matrix is not diagonalisable

2. Check if the following matrices are diagonalisable:

$$\begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \qquad \begin{pmatrix} 0 & -2 & -3 \\ 1 & 3 & 3 \\ 0 & 0 & 1 \end{pmatrix}.$$

② $\det \begin{bmatrix} -\lambda & -2 & -3 \\ 1 & 3-\lambda & 3 \\ 0 & 0 & 1-\lambda \end{bmatrix} = 0$

$\therefore \ 0 + 0 + (1-\lambda) \begin{vmatrix} -\lambda & -2 \\ 1 & 3-\lambda \end{vmatrix} = (1-\lambda)(\lambda^2 - 3\lambda + 2)$

$= (1-\lambda)(\lambda-1)(\lambda-2)$

$\therefore \ \lambda_1 = \lambda_2 = 1, \ \lambda_3 = 2$

For $\lambda_1 = \lambda_2 = 1$,

$\begin{bmatrix} -1 & -2 & -3 \\ 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix} u_{1,2} = 0 \quad \therefore \begin{bmatrix} -1 & -2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} u_{1,2} = 0$

$\therefore \ u_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} p, \ u_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} q$ $\quad \therefore$ Nullity $= 2 =$ number repeated $\lambda = 2$

$(p, q \neq 0)$

For $\lambda_3 = 2$,

$\begin{bmatrix} -2 & -2 & -3 \\ 1 & 1 & 3 \\ 0 & 0 & -1 \end{bmatrix} u_3 = 0 \quad \therefore \begin{bmatrix} -2 & -2 & -3 \\ 0 & 0 & 1.5 \\ 0 & 0 & -1 \end{bmatrix} u_3 = 0$

$\therefore \ u_3 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} t \ (t \neq 0)$ $\quad \therefore$ Nullity $= 1 =$ number repeated $\lambda = 1$

$\therefore$ Matrix is diagonalisable

3. Let $\mathbf{A} \in \mathbb{C}^{n\times n}$ be diagonalisable. Then, show

$$trace(\mathbf{A}) = \sum_{i=1}^{n} \lambda_i, \tag{1}$$

where λ_i is eigenvalue of $\mathbf{A}$. (Hint: use the definition of trace with the diagonalised form of $\mathbf{A}$).

Let $A = U \Lambda U^{-1}$,

$$\begin{aligned} \therefore \ trace(A) &= trace(U \Lambda U^{-1}) \\ &= trace(U U^{-1} \Lambda) \\ &= trace(I \Lambda) \\ &= trace(\Lambda) \\ &= \sum_{i=1}^{n} \lambda_i \end{aligned}$$

4. Consider a population of rabbits split into three age classes - those in their first year of life, those in their second year of life, and those in their third year of life. Further consider that:

• Half of the rabbits in their first year survive to their second year. Half of the rabbits in their second year survive to their third year. The maximum life span is 3 years.

• During the first year, the rabbits have no offspring. The average number of offspring is 6 during the second year and 8 during the third year.

(a) The population now consists of 24 rabbits in the first age class, 24 in the second, and 20 in the third. How many rabbits will there be in each age class in one years time?

(b) Find an age distribution which maintains the same ratio of the three age classes with growth of the population?

Hint: Find a matrix equation, $\mathbf{y} = \mathbf{A}\mathbf{x}$, where $\mathbf{x} = [x_1\ x_2\ x_3]^T$ (x_1 : the number of the 1-year-old rabbits, x_2: the number of the 2-year old rabbits, x_3: the number of the 3-year old rabbits) and $\mathbf{y} = [y_1\ y_2\ y_3]^T$ (y_1 : the number of the 1-year-old rabbits a year later, y_2: the number of the 2-year old rabbits a year later, y_3: the number of the 3-year old rabbits a year later).

(a) 1st age class : $24 \rightarrow 24 \times 6 + 20 \times 8 = 304$

2nd age class : $24 \rightarrow \frac{1}{2} \times 24 = 12$

3rd age class : $20 \rightarrow \frac{1}{2} \times 24 = 12$

(b) Following the Hint, $y = Ax = \begin{bmatrix} 0 & 6 & 8 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

For $y = Ax = \lambda x$,

$\det(A - \lambda I) = 0$

$$\therefore \det\begin{bmatrix} -\lambda & 6 & 8 \\ \frac{1}{2} & -\lambda & 0 \\ 0 & \frac{1}{2} & -\lambda \end{bmatrix}$$

$$= 0 - \frac{1}{2}\begin{vmatrix} -\lambda & 8 \\ \frac{1}{2} & 0 \end{vmatrix} - \lambda\begin{vmatrix} -\lambda & 6 \\ \frac{1}{2} & -\lambda \end{vmatrix} = 0 + 2 + (-\lambda^3 + 3\lambda)$$

$$= (\lambda - 2)(\lambda^2 + 2\lambda + 1) = (\lambda - 2)(\lambda + 1)^2$$

For $\lambda_1 = \lambda_2 = -1$,

$$\begin{bmatrix} 1 & 6 & 8 \\ \frac{1}{2} & 1 & 0 \\ 0 & \frac{1}{2} & 1 \end{bmatrix} u_{1,2} = 0 \qquad \therefore \begin{bmatrix} -1 & 6 & 8 \\ 0 & -2 & -4 \\ 0 & 0 & 0 \end{bmatrix} u_{1,2} = 0$$

$$\therefore u_{1,2} = \begin{bmatrix} -4 \\ -2 \\ 1 \end{bmatrix} t = \begin{bmatrix} 4 \\ 2 \\ -1 \end{bmatrix} t \; (t \neq 0)$$

For $\lambda_3 = 2$,

$$\begin{bmatrix} -2 & 6 & 8 \\ \frac{1}{2} & -2 & 0 \\ 0 & \frac{1}{2} & -2 \end{bmatrix} u_3 = 0 \qquad \therefore \begin{bmatrix} -2 & 6 & 8 \\ 0 & -\frac{1}{2} & 2 \\ 0 & 0 & 0 \end{bmatrix} u_3 = 0$$

$$\therefore u_3 = \begin{bmatrix} 16 \\ 4 \\ 1 \end{bmatrix} q \; (q \neq 0)$$

$$\because \lambda_{1,2} < 0$$

$\therefore$ x that maintain ratio $= \begin{bmatrix} 16 \\ 4 \\ 1 \end{bmatrix} q \; (q \neq 0)$

5. Show that $\mathbf{A}$ and $\mathbf{B}$ are not similar matrices:

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 4 \\ 0 & 2 & 3 \\ 0 & 0 & 4 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 4 & 0 \\ 2 & 3 & 4 \end{pmatrix}.$$

$\det(A) = 0 - 0 + 4 \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} = 16$

$\det(B) = 1 \begin{vmatrix} 4 & 0 \\ 3 & 4 \end{vmatrix} - 0 + 0 = 16$

$\therefore \det(A) = \det(B)$

$\text{trace}(A) = 2+2+4 = 8$

$\text{trace}(B) = 1+4+4 = 9$

$\therefore \text{trace}(A) \neq \text{trace}(B)$

$\therefore$ A and B are not simillar

6. Let $\mathbf{A}$ be

$$\begin{pmatrix} -2 & 4 \\ -1 & 3 \end{pmatrix}.$$

(a) find $\mathbf{A}^6$ by diagonalising $\mathbf{A}$ such that $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$.

(b) Now, using the Cayley-Hamilton theorem, find $\mathbf{A}^6$.

(a) $\det(A - \lambda I) = \det\begin{bmatrix} -2-\lambda & 4 \\ -1 & 3-\lambda \end{bmatrix} = (-2-\lambda)(3-\lambda) - (-4)$

$= \lambda^2 - 3\lambda + 2\lambda - 6 + 4 = \lambda^2 - \lambda - 2 = (\lambda-2)(\lambda+1) = 0$

$\therefore \lambda_1 = 2,\ \lambda_2 = -1$ $\qquad \therefore \Lambda = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$

For $\lambda_1 = 2$,

$\begin{bmatrix} -4 & 4 \\ -1 & 1 \end{bmatrix} u_1 = 0$ $\qquad \therefore \begin{bmatrix} -4 & 4 \\ 0 & 0 \end{bmatrix} u_1 = 0$

$\therefore u_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} p \ (p \neq 0)$

For $\lambda_2 = -1$,

$\begin{bmatrix} -1 & 4 \\ -1 & 4 \end{bmatrix} u_2 = 0$ $\qquad \therefore \begin{bmatrix} -1 & 4 \\ 0 & 0 \end{bmatrix} u_2 = 0$

$\therefore u_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix} q \ (q \neq 0)$

$\therefore X = \begin{bmatrix} p & 4q \\ p & q \end{bmatrix}$

$\therefore X^{-1} = \dfrac{\operatorname{adj}(X)}{\det(X)} = \dfrac{\begin{bmatrix} q & -p \\ -4q & p \end{bmatrix}^T}{pq - 4pq} = -\dfrac{1}{3pq}\begin{bmatrix} q & -4q \\ -p & p \end{bmatrix}$

$\therefore A^6 = X\Lambda^6 X^{-1} = -\dfrac{1}{3pq}\begin{bmatrix} p & 4q \\ p & q \end{bmatrix}\begin{bmatrix} 64 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} q & -4q \\ -p & p \end{bmatrix}$

6. Let $\mathbf{A}$ be

$$\begin{pmatrix} -2 & 4 \\ -1 & 3 \end{pmatrix}.$$

(a) find $\mathbf{A}^6$ by diagonalising $\mathbf{A}$ such that $\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1}$.

(b) Now, using the Cayley-Hamilton theorem, find $\mathbf{A}^6$.

$$\therefore A^6 = -\frac{1}{3pq}\begin{bmatrix} 64p & 4q \\ 64p & q \end{bmatrix}\begin{bmatrix} q & -4q \\ -p & p \end{bmatrix}$$

$$= -\frac{1}{3pq}\begin{bmatrix} 60pq & -252pq \\ 63pq & -255pq \end{bmatrix} = \begin{bmatrix} -20 & 84 \\ -21 & 85 \end{bmatrix}$$

(b) $P(\lambda) = \det(A - \lambda I) = \lambda^2 - \lambda - 2$

$\therefore P(A) = A^2 - A - 2I = 0$

$\therefore A^2 = A + 2I$

$\therefore A^6 = A^2 A^4 = A^5 + 2A^4 = A^2 A^3 + 2A^2 A^2$

$= A^4 + 2A^3 + 2A^2 + 8A + 8I$

$= (A^2 + 4A + 4I) + (2A^2 + 4A) + (2A + 4I) + (8A + 8I)$

$= (5A + 6I) + (6A + 4I) + (10A + 12I)$

$= 21A + 22I$

$$= \begin{bmatrix} -42 & 84 \\ -21 & 63 \end{bmatrix} + \begin{bmatrix} 22 & 0 \\ 0 & 22 \end{bmatrix} = \begin{bmatrix} -20 & 84 \\ -21 & 85 \end{bmatrix}$$

7. Let an inner product of two complex matrices $\mathbf{A}$ and $\mathbf{B}$ be defined by

$$\langle \mathbf{A}, \mathbf{B} \rangle = trace(\mathbf{B}^H \mathbf{A}).$$

If $\mathbf{A}$ and $\mathbf{B}$ are given by

$$\mathbf{A} = \begin{pmatrix} -2+i & 0 \\ 0 & 3-i \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 5-10i \\ -4+3i & 0 \end{pmatrix},$$

respectively, show $\mathbf{A}$ and $\mathbf{B}$ are orthogonal with respect to the given inner product.

$$< A, B > = \text{trace} < B^H A >$$

$$= \text{trace} < \begin{bmatrix} 0 & -4-3i \\ 5+10i & 0 \end{bmatrix} \begin{bmatrix} -2+i & 0 \\ 0 & 3-i \end{bmatrix} >$$

$$= \text{trace} < \begin{bmatrix} 0 & - \\ - & 0 \end{bmatrix} >$$

$$= 0$$

$\therefore$ A and B are orthogonal